

Module 4

Circuit Analysis in the S Domain

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Passive Filters

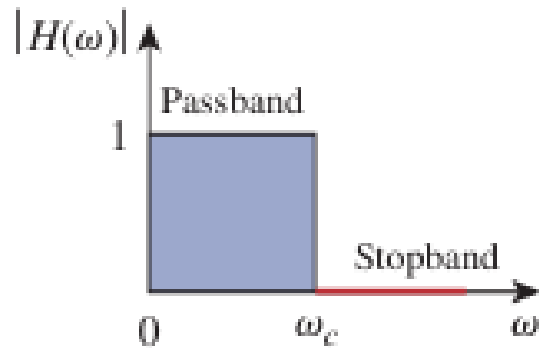
A filter is a circuit that is designed to pass signals with desired frequencies and reject or attenuate others. As a frequency-selective device, a filter can be used to limit the frequency spectrum of a signal to some specified band of frequencies. Filters are the circuits used in radio and TV receivers to allow us to select one desired signal out of a multitude of broadcast signals in the environment.

A filter is a passive filter if it consists of only passive elements R , L , and C . It is said to be an active filter if it consists of active elements (such as transistors and op amps) in addition to passive elements R , L , and C . The filter can be of any order.

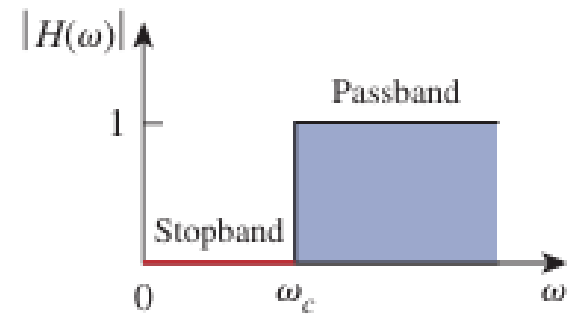
1. A lowpass filter passes low frequencies and stops high frequencies.
2. A highpass filter passes high frequencies and rejects low frequencies.
3. A bandpass filter passes frequencies within a frequency band and blocks or attenuates frequencies outside the band.
4. A bandstop filter passes frequencies outside a frequency band and blocks or attenuates frequencies within the band.

Passive Filters

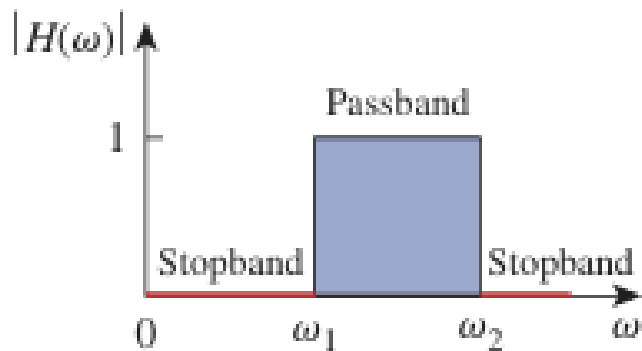
Lowpass filter



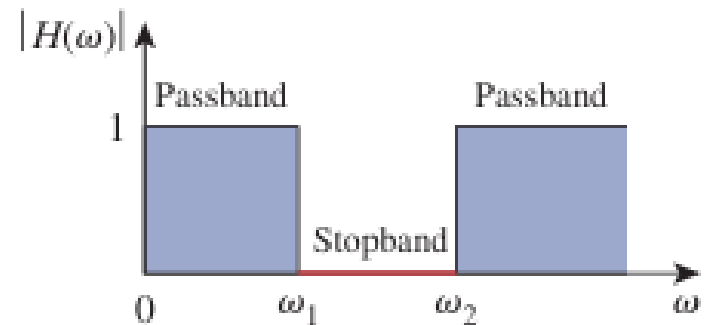
Highpass filter



Bandpass filter



Bandstop filter

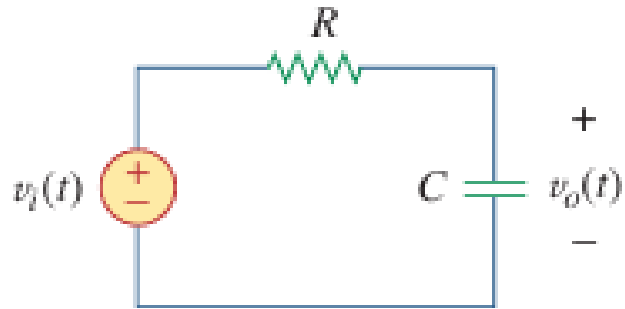


Characteristics of ideal Filters

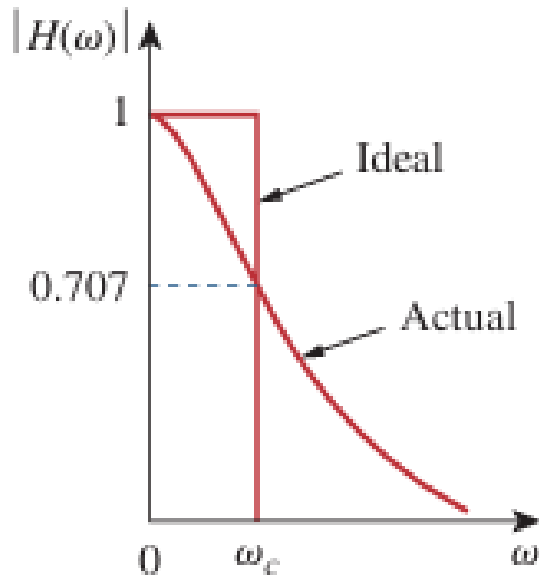
Type of Filter	$H(0)$	$H(\infty)$	$H(\omega_c)$ or $H(\omega_0)$
Lowpass	1	0	$1/\sqrt{2}$
Highpass	0	1	$1/\sqrt{2}$
Bandpass	0	0	1
Bandstop	1	1	0

ω_c is the cutoff frequency for lowpass and highpass filters; ω_0 is the center frequency for bandpass and bandstop filters.

Lowpass Filters



Frequency response:



$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{1/j\omega C}{R + 1/j\omega C}$$

$$\mathbf{H}(\omega) = \frac{1}{1 + j\omega RC}$$

Note that $H(0) = 1$ and $H(\infty) = 0$

The half-power frequency or the cutoff frequency of the filter is obtained by setting the magnitude of $H(\omega)$ equal to $1/\sqrt{2}$.

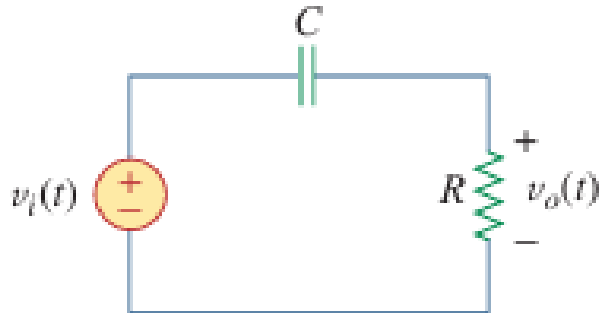
$$H(\omega_c) = \frac{1}{\sqrt{1 + \omega_c^2 R^2 C^2}} = \frac{1}{\sqrt{2}}$$

$$\omega_c = \frac{1}{RC}$$

The **cutoff frequency** is also called as the **roll-off frequency**. A lowpass filter is designed to pass only frequencies from dc up to the cutoff frequency ω_c .

The cutoff frequency is the frequency at which the transfer function $\mathbf{H}$ drops in magnitude to 70.71% of its maximum value. It is also regarded as the frequency at which the power dissipated in a circuit is half of its maximum value.

Highpass Filters

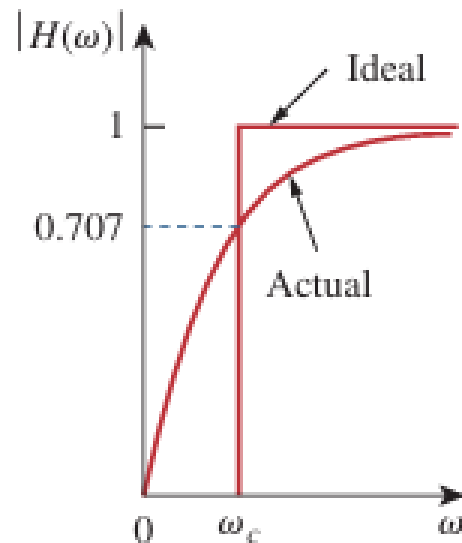


$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{R}{R + 1/j\omega C}$$

$$\mathbf{H}(\omega) = \frac{j\omega RC}{1 + j\omega RC}$$

Frequency response:

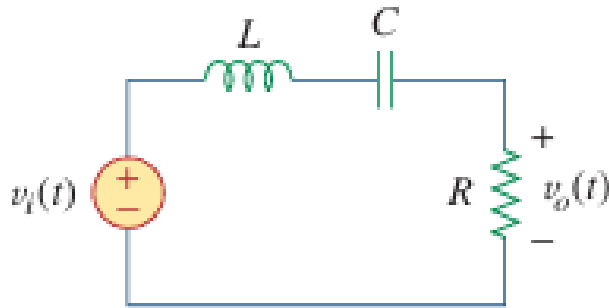
Note that $H(0) = 0$ and $H(\infty) = 1$



$$\omega_c = \frac{1}{RC}$$

A highpass filter is designed to pass all frequencies above its cutoff frequency ω_c .

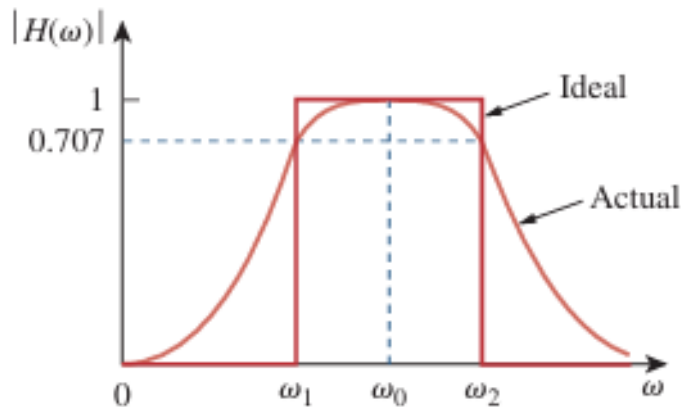
Bandpass Filters



$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{R}{R + j(\omega L - 1/\omega C)}$$

Note that $H(0) = 0$ and $H(\infty) = 0$

Frequency response:



The bandpass filter passes a band of frequencies $\omega_1 < \omega < \omega_2$ centered on ω_0 the centre frequency.

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

$$\omega_0 = \sqrt{\omega_1 \omega_2}$$

$$Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 C R}$$

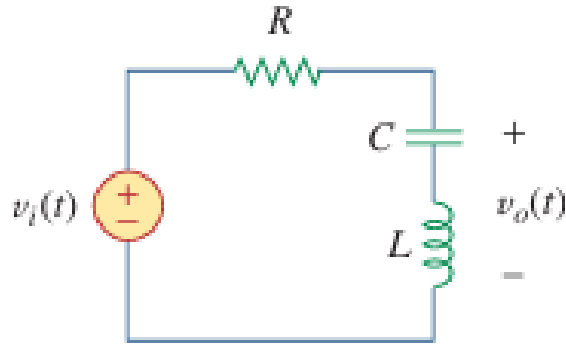
A bandpass filter can also be formed by cascading the lowpass filter (where $\omega_2 = \omega_c$) with highpass filter (where $\omega_1 = \omega_c$).

The bandpass filter is a series resonant circuit, the bandwidth, and the quality factor are related:

$$B = \frac{R}{L} = \frac{\omega_0}{Q}$$

$$B = \omega_0^2 C R.$$

Bandstop Filters

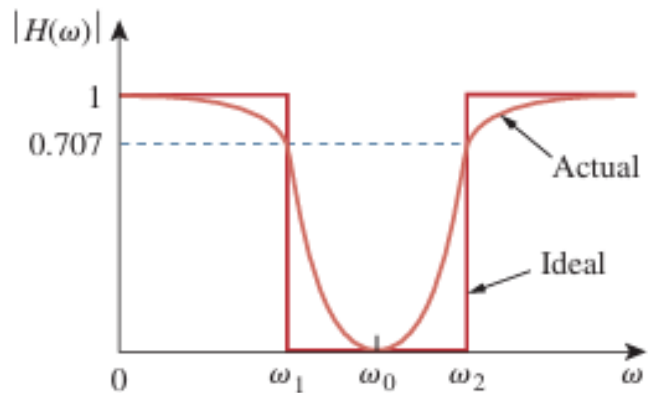


A filter that prevents a band of frequencies between two designated values (ω_1 and ω_2) from passing is known as a bandstop, band reject, or notch filter.

A bandstop filter is formed when the output RLC series resonant circuit is taken off the LC series combination.

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{j(\omega L - 1/\omega C)}{R + j(\omega L - 1/\omega C)}$$

Note that $H(0) = 1$ and $H(\infty) = 1$



$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Rejection frequency

$$\text{Bandwidth} = \omega_2 - \omega_1$$

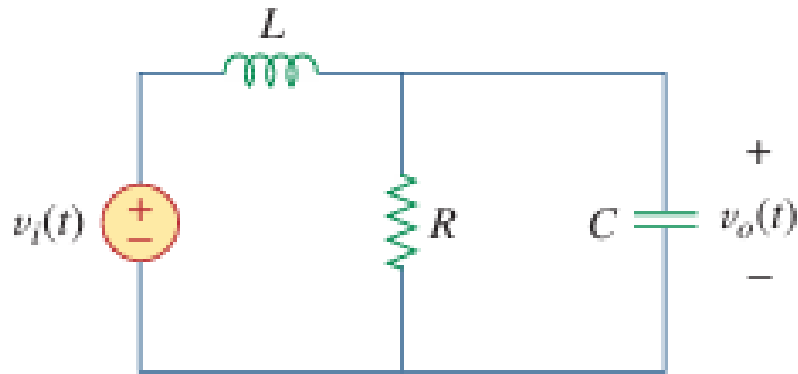
A bandstop filter is designed to stop or eliminate all frequencies within a band of frequencies, $\omega_1 < \omega < \omega_2$.

Passive Filters (Bandpass & Bandstop)

Resonant frequency, ω_0	$\frac{1}{\sqrt{LC}}$	$\frac{1}{\sqrt{LC}}$
Quality factor, Q	$\frac{\omega_0 L}{R}$ or $\frac{1}{\omega_0 RC}$	$\frac{R}{\omega_0 L}$ or $\omega_0 RC$
Bandwidth, B	$\frac{\omega_0}{Q}$	$\frac{\omega_0}{Q}$
Half-power frequencies, ω_1, ω_2	$\omega_0 \sqrt{1 + \left(\frac{1}{2Q}\right)^2} \pm \frac{\omega_0}{2Q}$	$\omega_0 \sqrt{1 + \left(\frac{1}{2Q}\right)^2} \pm \frac{\omega_0}{2Q}$
For $Q \geq 10, \omega_1, \omega_2$	$\omega_0 \pm \frac{B}{2}$	$\omega_0 \pm \frac{B}{2}$

Passive Filters

1. Determine what type of filter is shown below. Calculate the corner or cutoff frequency. Take $R = 2k$, $L = 2H$, $C = 2\mu F$.



$$\mathbf{H}(s) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{R \parallel 1/sC}{sL + R \parallel 1/sC}, \quad s = j\omega$$

But

$$R \parallel \frac{1}{sC} = \frac{R/sC}{R + 1/sC} = \frac{R}{1 + sRC}$$

$$\mathbf{H}(s) = \frac{R/(1 + sRC)}{sL + R/(1 + sRC)} = \frac{R}{s^2RLC + sL + R}, \quad s = j\omega$$

$$\mathbf{H}(\omega) = \frac{R}{-\omega^2RLC + j\omega L + R}$$

Since, $H(0) = 1$ and $H(\infty) = 0$, this is a low pass filter (second order).

Magnitude of $H(\omega)$ is,

$$H = \frac{R}{\sqrt{(R - \omega^2RLC)^2 + \omega^2L^2}}$$

Passive Filters

We know that the corner frequencies have $H = 1/\sqrt{2}$.

$$H = \frac{R}{\sqrt{(R - \omega^2 RLC)^2 + \omega^2 L^2}}$$

$$H^2 = \frac{1}{2} = \frac{R^2}{(R - \omega_c^2 RLC)^2 + \omega_c^2 L^2}$$

$$2 = (1 - \omega_c^2 LC)^2 + \left(\frac{\omega_c L}{R}\right)^2$$

Substituting the values of R , L , and C , we obtain

$$2 = (1 - \omega_c^2 4 \times 10^{-6})^2 + (\omega_c 10^{-3})^2$$

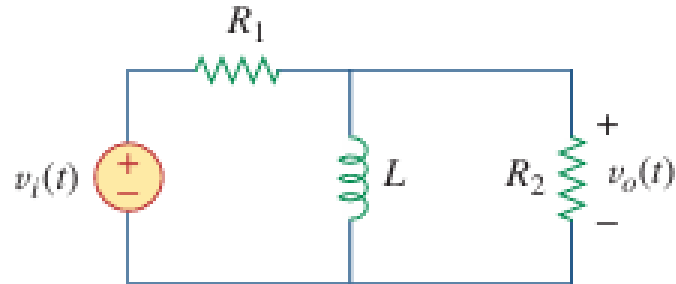
Assuming that ω_c is in krad/s,

$$2 = (1 - 4\omega_c^2)^2 + \omega_c^2 \quad \text{or} \quad 16\omega_c^4 - 7\omega_c^2 - 1 = 0$$

$$\omega_c = 0.742 \text{ krad/s} = 742 \text{ rad/s}$$

Passive Filters

2. For the circuit below, obtain the transfer function $V_o(\omega) / V_i(\omega)$. Identify the type of filter the circuit represents and determine the corner frequency. Take $R_1 = 100 \, \Omega = R_2$, $L = 2 \, \text{mH}$.



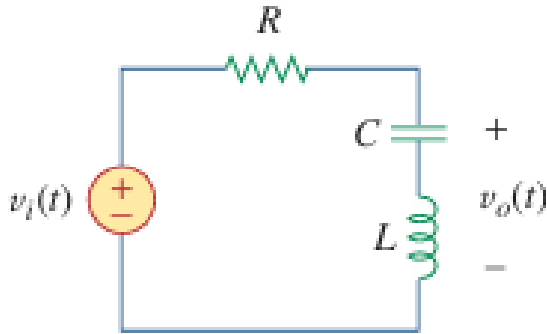
Answer:

$$\frac{R_2}{R_1 + R_2} \left(\frac{j\omega}{j\omega + \omega_c} \right), \text{ highpass filter}$$

$$\omega_c = \frac{R_1 R_2}{(R_1 + R_2)L} = 25 \text{ krad/s.}$$

Passive Filters

3. If the bandstop filter in figure shown below is to reject a 200-Hz sinusoid while passing other frequencies, calculate the values of L and C. Take $R = 150 \, \Omega$ and the bandwidth as 100 Hz.



Express bandwidth in radians/sec, $B = 2\pi(100) = 200\pi \text{ rad/s}$

But,

$$B = \frac{R}{L} \Rightarrow L = \frac{R}{B} = \frac{150}{200\pi} = 0.2387 \text{ H}$$

Rejection of the 200-Hz sinusoid means that f_0 is 200 Hz.

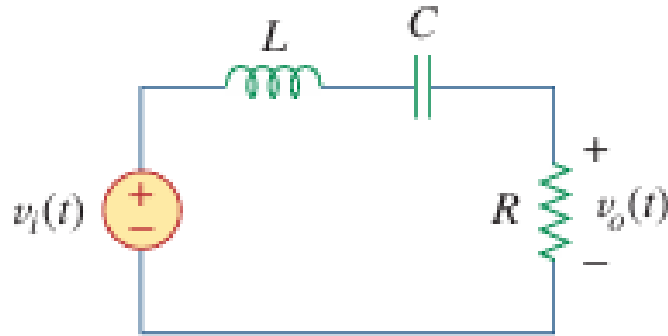
$$\omega_0 = 2\pi f_0 = 2\pi(200) = 400\pi$$

$$\omega_0 = 1/\sqrt{LC},$$

$$C = \frac{1}{\omega_0^2 L} = \frac{1}{(400\pi)^2 (0.2387)} = 2.653 \, \mu\text{F}$$

Passive Filters

4. Design a bandpass filter of the form shown in figure with a lower cutoff frequency of 20.1 kHz and an upper cutoff frequency of 20.3 kHz. Take $R = 20\text{k}\Omega$. Calculate L , C , and Q .



Answer:

15.915 H, 3.9 pF, 101.

Next class: Transfer function